

EEE4440 Power Electronics in Energy Systems

Group Project Assignment

Design and Analysis of a High-Gain SEPIC Converter for Photovoltaic Applications

Following your literature review pre-submission, where each group member investigated recent research on high-gain SEPIC converter topologies, you are now expected to complete the full converter design process as a group.

The objective of this project is to design a high-gain SEPIC DC–DC converter capable of stepping up the voltage of a selected photovoltaic (PV) source while satisfying given engineering requirements and constraints.

Your work must include converter selection justification, analytical design calculations, simulation, and analog control circuit design.

Input Source (PV Panel)

Select either a single PV panel or an appropriate PV panel array.

Design Requirement

Your converter must provide an output voltage equal to 5-6 x the open-circuit voltage (V_{oc}) of the selected PV panel.

Output current requirement is 2A.

Each group may independently choose:

- Switching frequency
- Semiconductor devices (MOSFET, diode)
- Passive components (inductors, capacitors)
- High-gain SEPIC variant/topology modifications
- Control strategy parameters

Design choices must be technically justified.

Required Project Components

1. Converter Topology Selection and Justification

Explain:

- Why SEPIC topology is suitable
- Why a high-gain topology is necessary
- Advantages compared to conventional boost converter

2. Design Calculations

Provide complete analytical calculations for:

- Duty cycle
- Inductor
- Capacitor
- Voltage/current on components
- MOSFET selection
- Diode selection

Clearly state all assumptions.

3. MATLAB/Simulink Model

Build a full simulation model including:

- PV source (first case) and a variable voltage source to demonstrate duty-cycle adjustment (second case)
- High-gain SEPIC converter
- PWM switching
- Output load
- Measurement blocks

Simulation should demonstrate:

- Correct output voltage
- Converter startup behavior
- Steady-state waveforms
- Voltage and current ripple

4. Analog Control Design

Design an analog controller for converter regulation.

- PWM controller implementation
- Gate drive strategy
- Feedback sensing

Complete PCB implementation is not required, but the control circuit design should be realistic and technically consistent.

5. Literature Integration

You must reference the papers collected during your **pre-submission**.

At least **two ideas or design decisions** in your final project should be supported by findings from your literature review.

Final Submissions

A. Final Report (PDF)

Include:

- Group number
- Student names and IDs
- Introduction
- Literature-based design motivation
- Converter calculations
- Component selection
- Simulation results
- Analog control design
- Discussion and conclusions
- References

B. MATLAB/Simulink Files

Upload:

- .slx model
- Any custom scripts used

C. Analog controller design files

Upload:

- Controller design/simulation files (PSpice, LTspice, or equivalent circuit simulation source files)

Presentations

(16th of June based on the registered schedule)

PowerPoint presentations are not required, we will use your final reports during the presentation.

Each group will briefly present:

- Design approach
- Key calculations
- Simulation results